4 **(a)** A source oscillates with frequency f to produce a progressive wave of wavelength λ. The source takes time t to produce n complete oscillations.

(i) State what is meant by a progressive wave.

....

.... [1]

(ii) State expressions, in terms of some or all of f, λ and n, for:

- the distance moved by a wavefront in time t

distance = ...

- time t.

time t = ...
[2]

(iii) Use your answers in **(ii)** to determine an expression for the speed v of the wave in terms of f and λ.

[1]

(b) Two identical microwave sources X and Y emit waves in phase. The sources are separated by a distance of 30 cm, as shown in Fig. 4.1.

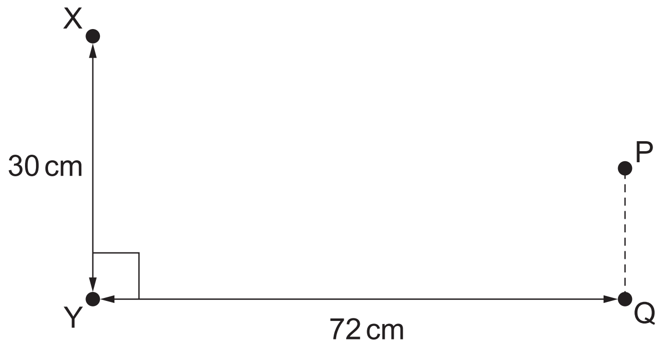

Fig. 4.1 (not to scale)

The intensity of the microwaves is to be investigated at points P and Q.
Line PQ is parallel to line XY. Distance XP is equal to distance YP. Distance YQ is 72 cm and angle XYQ is 90°.
The wavelength of the microwaves is 4.0 cm.